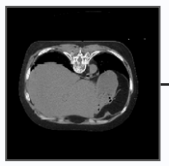
A PLC-Net generator (2-D U-Net)



NCCT x
1 × 128 × 128
HU [–200, 400] → [0, 1]

2.5-D variant: 11 adjacent slices as input channels (in = 11)

enc1 1 → 64 128²
[Conv3×3 → GN → LReLU] ×2

MaxPool 2×2

enc2 64 → 128 64²
[Conv3×3 → GN → LReLU] ×2

MaxPool 2×2

enc3 128 → 256 32²
[Conv3×3 → GN → LReLU] ×2

MaxPool 2×2

enc4 256 → 512 16²
[Conv3×3 → GN → LReLU] ×2

MaxPool 2×2

Bottleneck 512 → 512 8²
[Conv3×3 → GN → LReLU → Drop] ×2

Encoder is unconditioned: shared across phases & levels

copy & concat

copy & concat

copy & concat

copy & concat

dec4 (256+512) → 256 16²
Conv3×3 → GN → LReLU → Drop
Conv3×3 → GN → LReLU

dec3 (128+256) → 128 32²
Conv3×3 → GN → LReLU → Drop
Conv3×3 → GN → LReLU

dec2 (64+128) → 64 64²
Conv3×3 → GN → LReLU → Drop
Conv3×3 → GN → LReLU

dec1 (64+64) → 64 128²
Conv3×3 → GN → LReLU → Drop
Conv3×3 → GN → LReLU

ConvT 2×2, s2 · 512 → 256

ConvT 2×2, s2 · 256 → 128

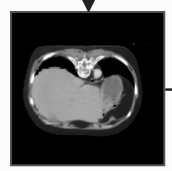
ConvT 2×2, s2 · 128 → 64

ConvT 2×2, s2 · 64 → 64

FiLM 1 **FiLM 2** **FiLM 3** **FiLM 4** **FiLM 5**

γ, β

Conv1×1
64 → 1
→ Sigmoid



Synthetic CECT ŷ
1 × 128 × 128

ŷ

GN = GroupNorm (8 groups) · **LReLU** = LeakyReLU(0.2) · **Drop** = Dropout(0.2) · **FiLM**: y = x·(1+γ(c)) + β(c)

c (detached: no gradient to G)

B Conditioning branch (decoder only)

Phase id p
venous = 0 · arterial = 1

nn.Embedding(2, 64)

phase emb.

Target organ levels z ∈ ℝ⁸
aorta
portal / splenic vein
inferior vena cava
heart
liver
pancreas
gallbladder
colon

Linear(8 → 64)
zero-init

level emb.

+

c ∈ ℝ⁶⁴

FiLM MLP (one per site, ×5)
Linear(64 → h) → ReLU → Linear(h → 2C)
last layer zero-init → (γ, β) ∈ ℝ^C × ℝ^C

FiLM
y = x · (1 + γ) + β
γ = β = 0 at init
⇒ identity at step 0

site	after	C	h
FiLM 1	bottleneck	512	128
FiLM 2	dec4	256	64
FiLM 3	dec3	128	64
FiLM 4	dec2	64	64
FiLM 5	dec1	64	64

z = median HU of each organ in the target CECT (TotalSegmentator mask), z-scored with train-split statistics pooled over phases; missing organ → 0

Training: target-derived level z from the real CECT

Main evaluation: pooled-reference level
Alternative: phase-specific reference level

C Conditional PatchGAN D

training only

NCCT x

real CECT y

concat [x , y or ŷ] 2 × 128²
D(x, ·) judges the pair, not ŷ alone

Conv4×4, s2 2 → 64 → LReLU
no norm · 64²

Conv4×4, s2 64 → 128 → BN → LReLU
32²

Conv4×4, s2 128 → 256 → BN → LReLU
16²

Conv4×4, s1 256 → 512 → BN → LReLU
15²

Linear
(64 → 512)
zero-init

bias

+

Conv4×4, s1 512 → 1
14 × 14 patch logit map (real / fake)

L_{LSGAN}

feature matching

D Training objective

TotalSegmentator organ mask M
training only

Target CECT y
(venous or arterial)

$\mathbf{L_G} = \lambda_{L1}(t) \cdot \|\hat{y} - y\|_1 + 20 \cdot L_{organ} + \lambda_{adv}(t) \cdot L_{LSGAN} + 10 \cdot L_{FM}$

$\mathbf{L_D} = ½ [(D(x, y) - 1)^2 + D(x, \hat{y})^2]$ (LSGAN)

- $\lambda_{L1}(t)$: 100 → 25, linear decay over epochs 10–30
- L_{organ}: organ-weighted L1, weights from M — aorta / IVC / portal v. 6, heart & great vessels 4, liver 3, other solid organs 2, background 0.5, lung / bone / muscle 0.25, GI tract 0
- $\lambda_{adv}(t) = 0.5$, linear warm-up over 5 epochs
- L_{FM}: L1 between D features of real and synthetic pairs (4 blocks)
- Adam (β = 0.5, 0.999), lr_G $2 \cdot 10^{-4}$, lr_D $1 \cdot 10^{-4}$, batch 16, ≤ 70 epochs (early stop, patience 30)